BEIT-BASED ALZHEIMER MRI IMAGE CLASSIFICATION PIPELINE: LEVERAGING TRANSFORMERS FOR ACCURATE DIAGNOSIS

Project Submitted to the
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submitted by

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May 2024

DECLARATION

I undersigned hereby declare that the project report **BEiT-based Alzheimer MRI Image Classification Pipeline: Leveraging Transformers for Accurate Diagnosis** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Dr. Nunna Satya Krishna. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

This is to certify that the report entitled **BEiT-based Alzheimer MRI Image Classification Pipeline: Leveraging Transformers for Accurate Diagnosis** submitted by **Ruhi Kashmi Tofia, Gurram Rishitha, Lokeshchandra Mandadapu, Hemanth Maddi** to the SRM University-AP in partial fulfillment of the requirements for the award of the Degree of Master of Technology in in is a bonafide record of the project work carried out under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Signature:	Signature: .....

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I wish to express my profound gratitude and indebtedness to all those who contributed to the preparation of this Project Report titled **BEiT-based Alzheimer MRI Image Classification Pipeline: Leveraging Transformers for Accurate Diagnosis** and helped me present it satisfactorily. First and foremost, I am immensely grateful to God for guiding me throughout this journey. Special thanks are due to my mentor and supervisor, Dr. Nunna Satya Krishna from the Computer Science & Engineering , whose valuable suggestions and critical inputs significantly enriched this report. Additionally, I extend my gratitude to Prof. Niraj Upadhyaya, the Head of the Computer Science & Engineering for his encouragement and support.

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ABSTRACT

Alzheimer's disease (AD), a neuro-degenerative condition, affects millions of individuals worldwide. The key to successful therapy and intervention for Alzheimer's disease is early and precise diagnosis. Because it may detect structural brain alterations, magnetic resonance imaging (MRI) is a common diagnostic method for Alzheimer's disease (AD). Using the Bidirectional Encoder representation from Image Transformers (BEiT) model, we provide a deep learning strategy for Alzheimer's disease MRI image classification into distinct phases. In order to train the model efficiently, the pipeline pre-processes the MRI dataset, divides it into training, validation, and test sets, and creates data loaders. Our model of choice is the BEiT, a cutting-edge architecture that was pre-trained on massive picture datasets using transformers. The number of classes in the Alzheimer's disease dataset is adjusted in the final BEiT model classification layer. During the training of the model, Taking into consideration the class imbalance through to the loss function. The Adam optimizer is used to train the model, and measures like F1-score, recall, accuracy, and precision are used to assess its performance.

CONTENTS

ACKNOWLEDGMENT **i**

ABSTRACT **ii**

LIST OF TABLES **iv**

LIST OF FIGURES **v**

Chapter 1. INTRODUCTION TO THE PROJECT **1**

1.1 Understanding Alzheimer's Disease 1

1.2 Importance of Early Detection 2

1.3 Role of Deep learning in Alzheimer's Detection 3

Chapter 2. MOTIVATION **5**

2.1 Personal Relevance and Empathetic Engagement 5

2.2 Factors Influencing Alzheimer's Disease 6

2.3 Escalating Prevalence of Alzheimer's Disease. 7

Chapter 3. LITERATURE SURVEY **9**

Chapter 4. DESIGN AND METHODOLOGY **10**

4.1 SYSTEM ARCHITECTURE 10

4.2 METHODOLOGY 10

Chapter 5. IMPLEMENTATION **14**

5.1 Data Preparation Module: 14

5.2 Data Loader Module: 15

5.3 Model Architecture Module: 15

5.4 Training Module: 18
5.5 Evaluation and Metrics Module: 20

Chapter 6. SOFTWARE TOOLS USED **22**

Chapter 7. RESULTS & DISCUSSION **24**

Chapter 8. CONCLUSION **26**
8.1 Scope of future work 27

REFERENCES **28**

LIST OF TABLES

7.1 Results 24

LIST OF FIGURES

2.1 Motivation. 5
2.2 Age vs Alzheimers 7
2.3 Prevalance of Alzheimer's Disease 8

4.1 System Architecture 10

5.1 BeiT Architecture 18

6.1 Python IDE 22
6.2 Prediction 23

7.1 Performance Metrics Graph 24
7.2 Confusion Matrix of BEiT 25

Chapter 1

INTRODUCTION TO THE PROJECT

1.1 UNDERSTANDING ALZHEIMER'S DISEASE

Alzheimer's ailment (AD) is a multifaceted and devastating neurodegenerative disorder that profoundly affects individuals, families, and societies worldwide. Defined by advancing cognitive deterioration, memory decline, and alterations in behavior, AD robs individuals of their independence, dignity, and quality of life. As the significant common cause of memory loss, AD presents a significant and growing global health challenge, with an estimated 50 million people currently living with the disease globally, a number projected to triple by 2050 if effective treatments are not found.

The impact of Alzheimer's disease extends far beyond those directly affected, placing an immense burden on caregivers, families, and healthcare systems. Caregivers often face emotional, physical, and financial strain as they provide round-the-clock care and support for their loved ones with AD. Family members experience grief and frustration as they witness the gradual decline of their loved one's cognitive abilities and independence. Healthcare systems struggle to meet the complex needs of individuals with AD, from diagnosis and treatment to long-term care and support services.

The economic burden of Alzheimer's disease is staggering, with expenses linked to healthcare, extended care, and decreased productivity reaching hundreds of billions of dollars annually. These costs are expected to

escalate dramatically as the global population ages, highlighting the urgent need for effective strategies to prevent, diagnose, and treat this disease.

In spite of years of investigation, Alzheimer's ailment persists incurable, with available treatments providing only modest symptomatic relief. Detecting AD in its initial stages is essential for maximizing treatment results and enhancing the well-being of those impacted. Interventions aimed at slowing disease progression and preserving cognitive function are most effective when initiated in the beginning stages of the AD. Therefore, there is an urgent need for reliable and accessible methods for early identification and diagnosis of AD.

1.2 IMPORTANCE OF EARLY DETECTION

Initial identification of Alzheimer's disease is paramount for several reasons, each contributing to improved outcomes and quality of life for individuals affected by the disease.

Firstly, early detection facilitates the timely initiation of interventions and treatments aimed at slowing disease progression and alleviating symptoms. Pharmacological interventions, such as cholinesterase inhibitors and memantine, have been shown to be more effective when administered in the initial phases of AD. These medications could help stabilize cognitive function, manage behavioral indications, and enhance the overall quality of life for individuals with AD. Furthermore, non-pharmacological approaches, such as cognitive stimulation therapy, physical exercise, and dietary modifications, are most beneficial when implemented early. These interventions can help preserve cognitive function, maintain independence, and delay the onset of more severe symptoms.

Secondly, initial diagnosis offers individuals and their families the

chance to strategize for the future and make well-informed choices regarding care.. By knowing about the diagnosis early on, individuals can participate in advance care planning, including legal and financial arrangements, healthcare preferences, and end-of-life care decisions. This proactive approach allows individuals to maintain control over their care and ensures that their wishes are respected as the disease progresses. Families also benefit from early diagnosis by having more time to adjust to the diagnosis, seek out resources and support services, and develop coping strategies for managing the challenges of care giving.

Moreover, early detection enables researchers to gain a deeper comprehension of the underlying mechanisms of AD and build more efficient therapies. By identifying individuals with initial-stage AD, researchers can examine the progression of the disease over time, identify biomarkers and genetic risk factors, and test novel therapeutic interventions. This knowledge is crucial for furthering our comprehension of AD and developing targeted treatments that address the underlying pathology of the disease. Early diagnosis also facilitates participation in clinical trials, allowing individuals to contribute to research efforts aimed at finding a cure or disease-modifying treatments for Alzheimer's disease.

1.3 ROLE OF DEEP LEARNING IN ALZHEIMER'S DETECTION

Recent advancements in deep learning have revolutionized medical diagnostics, particularly in the field of neuroimaging. Magnetic resonance imaging (MRI) is a powerful tool for visualizing structural changes in the brain associated with Alzheimer's disease (AD).Deep learning models leverage vast amounts of data and computational power to excel in tasks such as

image recognition and nlp. When applied to AD, these models hold great promise for automating image analysis and classification, thus empowering physicians to make informed decisions efficiently. Originally designed for natural language processing tasks, transformer-based models have recently emerged as a breakthrough in deep learning architectures. These models, including the Bidirectional Encoder representation from Image Transformers (BEiT), exhibit exceptional performance across various computer vision applications by leveraging their ability to capture intricate patterns and connections within sequential data.

This Project aims to leverage deep learning and transformer-based architectures, particularly BEiT, to classify MRI images for diagnosing Alzheimer's disease. The objective is to develop a reliable model capable of distinguishing between different stages of AD progression by extracting valuable information from MRI images. Such a model holds the potential to significantly benefit healthcare providers by delivering accurate and timely insights into the trajectory and prognosis of AD.

Chapter 2

MOTIVATION

2.1 PERSONAL RELEVANCE AND EMPATHETIC ENGAGEMENT

At the heart of why we chose to study AD classification is a strong personal connection to seeing how neurodegenerative disorders affect people and families we care about. Seeing the tough times and challenges faced by loved ones dealing with AD gives us a deep feeling of understanding and a strong promise to help in meaningful ways. By diving into studying AD types, we really want to respect the experiences of those dealing with the disease and work hard to make their lives better through learning and coming up with new ideas in science.

Figure 2.1: Motivation.

2.2 FACTORS INFLUENCING ALZHEIMER'S DISEASE

Understanding the different components that influence the evolution and advancement of Alzheimer's ailment is crucial for effective management and prevention strategies. This section explores the key determinants that contribute to Alzheimer's risk and progression, encompassing age, genetics, lifestyle factors, environmental exposures, medical conditions, socioeconomic factors, cognitive and brain reserve, as well as stress and mental health.

As people get older, they are more likely to develop Alzheimer's disease. This is important to know because it means we need to find ways to detect and treat it early, especially since more and more people around the world are getting older. Genetics also have a big impact on Alzheimer's disease. Some traits that people inherit from their parents can make them more likely to get the disease. Researchers are working to understand these genetic factors better so they can find ways to help people who are at risk.Lifestyle choices such as diet, exercise, and social engagement can impact the risk and advancement of Alzheimer's disease. Understanding the impact of lifestyle factors empowers individuals to adopt healthy behaviors that promote brain health and mitigate Alzheimer's risk. Exposure to environmental factors, such as air pollution or toxins, may contribute to the development of AD. Investigating the role of environmental exposures aids in analyzing disease etiology and recognizing potential preventive measures.

Certain medical conditions, such as cardiovascular disease or diabetes, may increase the risk of Alzheimer's disease. Recognizing the interconnections between Alzheimer's and other health conditions informs holistic approaches to disease prevention and management. Socioeconomic

factors, including education level, income, and access to healthcare, can influence Alzheimer's risk and outcomes. Addressing disparities in healthcare access and socioeconomic status is essential for equitable Alzheimer's prevention and care.

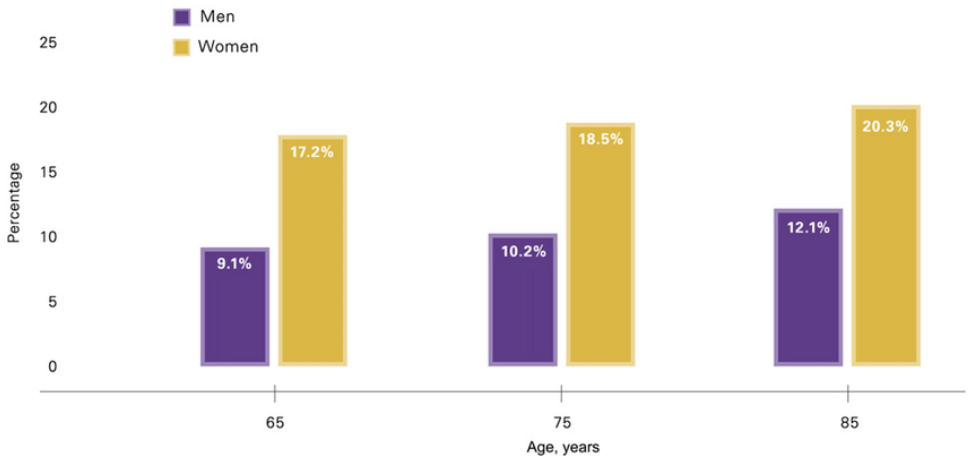


Figure 2.2: Age vs Alzheimers

2.3 ESCALATING PREVALENCE OF ALZHEIMER'S DISEASE.

Alzheimer's disease is becoming more common in today's world, and experts predict that this trend will continue in the future. By the year 2050, it's estimated that a huge 106.23 million people worldwide will have Alzheimer's disease.As people live longer and get older, more of them are getting Alzheimer's disease. This is because the risk of getting Alzheimer's increases as you get older. Also, changes in the way we live, like what we eat, how much we exercise, and what we're exposed to in the environment, can also play a part in why more people are getting Alzheimer's.Having more people with Alzheimer's disease has big effects on families, caregivers, and the community. Taking care of someone with Alzheimer's takes a lot of time and money. It also affects the economy because of the costs of healthcare and people not being able to work as much.

To deal with more people getting Alzheimer's, we need to work together. This means making sure people know how to keep their brains healthy, finding ways to spot Alzheimer's early, and doing more research to find treatments and ways to prevent Alzheimer's.

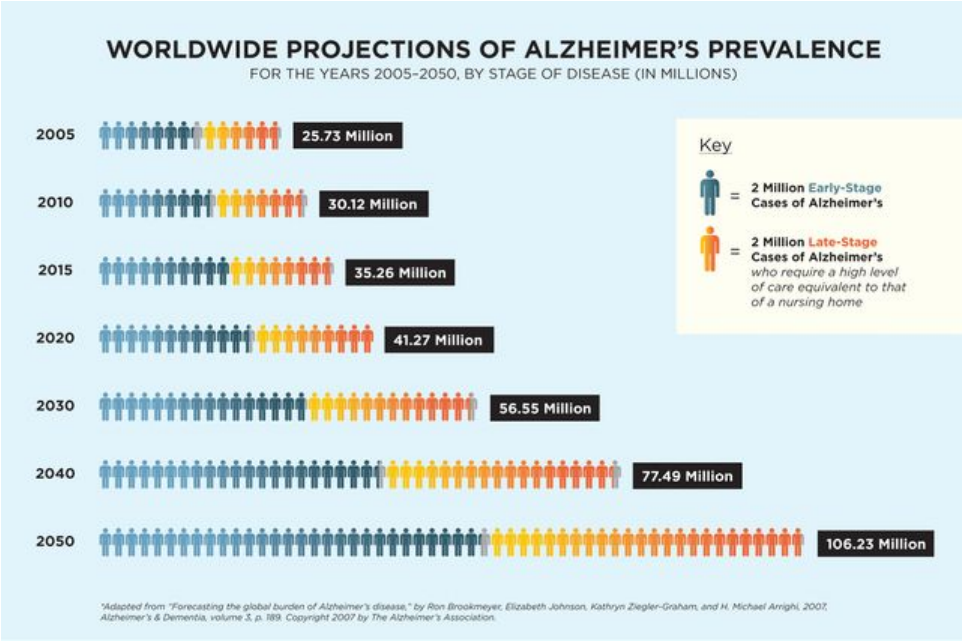


Figure 2.3: Prevalance of Alzheimer's Disease

Chapter 3

LITERATURE SURVEY

The literature survey provides a comprehensive overview of recent advancements in Diagnosing Alzheimer's disease through deep learning methodologies applied to MRI scans. Researchers have explored various deep learning architectures, involving convolutional neural networks (CNNs) [1], self-attention learners, and transformer-based models, to improve classification performance and contribute significantly to advancements in Alzheimer's disease diagnosis [2].

Research has underscored the significance of deep learning techniques in accurately analyzing MRI data for diagnosis. Additionally, comprehensive evaluations of existing approaches have underscored avenues for further research and development in this critical area of healthcare, emphasizing the potential of deep learning algorithms.

Novel approaches, such as cascaded multi-modal mixing transformers and hybrid architectures combining CNNs with attention mechanisms [3], have been proposed to address challenges such as incomplete data and improve classification accuracy [4].

Furthermore, classification frameworks based on deep learning techniques have been developed to aid in understanding disease progression and facilitating tailored treatment strategies effectively [5]. These advancements represent significant progress in medical image analysis, paving the way for more reliable diagnostic tools in clinical practice and facilitating early diagnosis and intervention for Alzheimer's disease [6].

Chapter 4

DESIGN AND METHODOLOGY

4.1 SYSTEM ARCHITECTURE

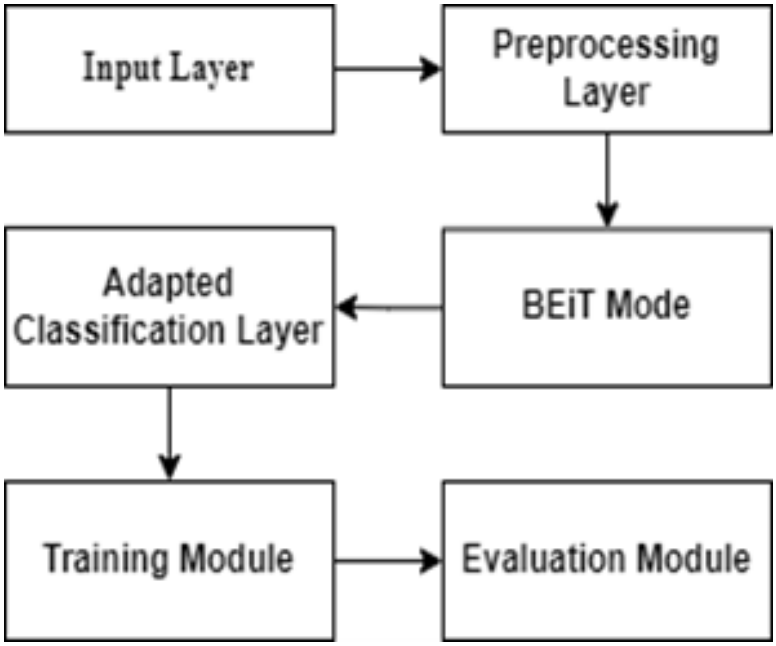


Figure 4.1: System Architecture

4.2 METHODOLOGY

The suggested method is designed to use deep learning techniques, more especially the BEiT model, to categorize magnetic resonance imaging (MRI) scans for the purpose of diagnosing Alzheimer's disease. There are several essential parts to this system. Resizing, normalizing, and augmenting are part of the preprocessing stages that MRI scans go through to make

sure the data is consistent and of high quality. The next step is to use the BEiT model for feature extraction and classification;BEIT encodes a given input image (x) into contextualized vector representations. The masked image modeling (MIM) task pretrained BEIT in a self-supervised learning method, as Figure 5.1 illustrates. Using encoding vectors as a basis, MIM seeks to retrieve the masked image patches. We add task layers on top of pretrained BEIT for downstream tasks (such semantic segmentation and picture classification) and adjust the parameters based on the particular datasets. The last classification layer of the BEiT model is adjusted to fit the precise number of classes that correspond to the various phases of Alzheimer's disease. When training, supervised learning methods are used in conjunction with optimization algorithms such as the Adam optimizer. Class weights are included into the loss function to improve the model's generalizability across various illness phases, which helps to address the class imbalance problems that are common in medical datasets. The trained model is then tested on a different validation dataset to ensure accurate assessment of its classification performance using measures like recall, accuracy, precision, and F1-score. Once the model's validation is complete and acceptable, it may be integrated into an existing healthcare infrastructure to help with Alzheimer's disease diagnosis and progression tracking. It can then be used as part of a clinical decision support system.

In this project, we employed a combination of libraries and tools in Python for data preprocessing, model development, evaluation, and deployment. The following is an overview of the methodologies and tools used:

- PyTorch and TorchVision: Utilized for constructing and training deep learning models, incorporating convolutional neural networks (CNNs),

and for handling image data. TorchVision was used for loading and transforming image datasets.

- Transforms: Used to preprocess images, including resizing and converting them into tensors.

- Train-Validation-Test Split: Split the dataset into training, validation,and test sets utilizing the train_test_split function from the sklearn.model_selection module.

- DataLoaders: Created data loaders using PyTorch's DataLoader class to efficiently load and iterate over batches of data during training, validation, and testing. Class Weights: Calculated class weights to address class imbalance in the dataset.

- Model Training: Implemented the training loop to train the deep learning model using PyTorch. Utilized an optimizer (e.g., AdamW) to update the model parameters based on the computed gradients. Employed We used a loss function, for example, cross-entropy loss, to assess the model's performance during training.

- Model Evaluation: Evaluated the trained model was evaluated using a validation set to monitor its performance and prevent overfitting. We calculated accuracy and other pertinent metrics to evaluate the model's performance on both the validation and test sets.

- Confusion Matrix and Classification Report: Generated a confusion matrix and a classification report using the confusion_matrix and classification_report functions from the sklearn.metrics_module to evaluate the model's performance in detail.

- Model Deployment: Saved the trained model's weights using PyTorch's torch.save function for future use. Utilized Flask, a lightweight web framework for Python, for building a web application to deploy the trained model. Serialized and deserialized Python objects using the pickle module, commonly used for saving and loading trained machine learning models.

- Other Libraries: Leveraged various other Python libraries for data manipulation, visualization, and analysis, including NumPy, Pandas, Matplotlib, and Seaborn. Used scikit-learn for additional machine learning tasks such as data preprocessing, feature transformation, and evaluation metrics. Employed TensorFlow and Keras for deep learning tasks, including building and training neural networks. Integrated regular expressions (re) for pattern matching and string manipulation tasks. Applied datetime for working with dates and times in Python.

By leveraging these methodologies and tools, we were able to develop, evaluate, and deploy an image classification model for Alzheimer's disease detection effectively.

Chapter 5

IMPLEMENTATION

5.1 DATA PREPARATION MODULE:

In this module focused on dataset preparation for training and evaluation, the process initiates with downloading the Alzheimer's 4-class dataset from Kaggle utilizing the kaggle library, followed by unzipping it. The Alzheimer's 4-class dataset comprises a total of 5,400 files, with each file representing an MRI scan. The dataset is categorized into four classes: Mild Demented (896 files), Moderate Demented (64 files), Non-Demented (3,200 files), and Very Mild Demented (2,240 files). This step ensures seamless access to the dataset required for training the deep learning model.

Subsequently, the module undertakes crucial image preprocessing operations to optimize the dataset for subsequent training and evaluation stages. These operations encompass resizing and transforming the images, facilitated by libraries such as torchvision and transforms. By applying resizing techniques, the images are standardized to a uniform size, ensuring consistency across the dataset. Additionally, transformations are employed to further enhance the quality and suitability of the images for input into the deep learning model. These transformations may include normalization, rotation, flipping, and cropping, among others, aimed at augmenting the dataset and enhancing its diversity, which can improve the model's generalization capabilities.

5.2 DATA LOADER MODULE:

The main goal of the Data Loader Module is to manage the effective loading of data batches throughout the assessment and training of the model. With the help of the torch, this is achieved.The DataLoader class is an essential part of the data handling tools in PyTorch. This class is used to generate data loaders for the training, validation, and test sets. This allows for mini-batches of iteration over the dataset to be done smoothly. When training or evaluating a model, the data loaders are vital in maximizing memory use and computing performance. Instead of processing the whole dataset all at once, the data loaders allow the model to make effective use of memory resources by loading data in mini-batches. Both the memory cost and the model's ability to handle big datasets that may not fit totally into memory are minimized by this.

In addition, by enabling the parallel processing of several data samples concurrently, mini-batches improve computing performance. This shortens the time it takes for the model to train and converge on an ideal answer. The Data Loader Module also makes it easy to split the dataset into different subsets for training and testing the model, so that each part is used correctly. If we want to know how well the model does on new data and avoid overfitting, this section is crucial. As a whole, the Data Loader Module is critical for improving data handling and facilitating mini-batching iterations over the dataset for training and evaluating models.

5.3 MODEL ARCHITECTURE MODULE:

In the Model Architecture Module, the foundation for the deep learning model tailored to Alzheimer's disease classification using MRI images is

laid. This begins with the selection and loading of a pre-trained BEiT (Bidirectional Encoder representation from Image Transformers) model from the transformers library. These pretrained models have undergone extensive training on vast image datasets, capturing intricate patterns and representations crucial for image classification tasks. By integrating a pretrained BEiT model, researchers harness the wealth of knowledge embedded within its parameters, enabling faster convergence and improved performance.

Image patches: The 2-dimensional image is divided into a series of patches so that image data can be directly input into a conventional Transformer. We reshape the image $x \in \mathbb{R}^{H \times W \times C}$ into $N = \frac{HW}{P^2}$ patches $x^p \in \mathbb{R}^{N \times (P^2 C)}$, where C is the number of channels, (H, W) is the input image resolution, and (P, P) is the resolution of each patch. Similar to word embeddings in BERT, the image patches are flattened into vectors and linearly projected. Image patches are utilized as input features in BEIT and retain the original pixel structure.

In our tests, we resize every 178×208 picture to 224×224. Then, we further divide each 224×224 image into a 14×14 grid of patches, each of size 16×16.

Visual tokens: Following the methodology described in RPG+21, we implement the image tokenizer acquired through discrete variational autoencoder (dVAE) training. This process involves two main modules: the tokenizer and the decoder. The tokenizer, denoted as $q_\psi(z|x)$, transforms image pixels x into discrete tokens z based on a visual codebook, essentially a vocabulary. Meanwhile, the decoder, denoted as $p_\psi(x|z)$, is responsible for reconstructing the input image x from the visual tokens z. The objective function for reconstruction can be expressed as $\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\psi(x|z)]$. Each image is tokenized into a 14×14 grid of visual tokens. It's worth mention-

ing that the quantity of visual tokens corresponds to the number of image patches for a single image.

The next crucial step involves adapting the final classification layer of the BEiT model to suit the specific requirements of Alzheimer's disease classification. Since the number of classes in the target dataset typically varies based on disease stages or diagnostic categories, the classification layer is modified accordingly. This adjustment ensures that the model's output aligns with the classification schema, facilitating accurate predictions of Alzheimer's disease status based on MRI images.

Furthermore, fine-tuning and optimization techniques are applied,refining the model to boost its performance. Fine-tuning involves updating the parameters of the pretrained BEiT model using the target MRI dataset, allowing the model to learn discriminative features relevant to Alzheimer's disease classification. Optimization strategies, such as gradient descent algorithms and learning rate scheduling, are employed to optimize model parameters and prevent overfitting, thereby improving generalization to unseen data.

Following model configuration, rigorous evaluation and validation procedures are conducted to assess its efficacy. The data is partitioned into training, validation, and testing subsets, ensuring robust evaluation metrics are calculated. Through iterative refinement based on evaluation results, researchers fine-tune the model architecture to achieve optimal performance in Alzheimer's disease classification. Ultimately, the Model Architecture Module serves as the cornerstone for developing a robust deep learning framework capable of accurately classifying Alzheimer's disease using MRI images. By leveraging pretrained models, adapting the classification layer, and optimizing model parameters, researchers pave the way for reliable and efficient diagnosis, crucial for timely strategies for intervention and

treatment planning in Alzheimer's disease management.

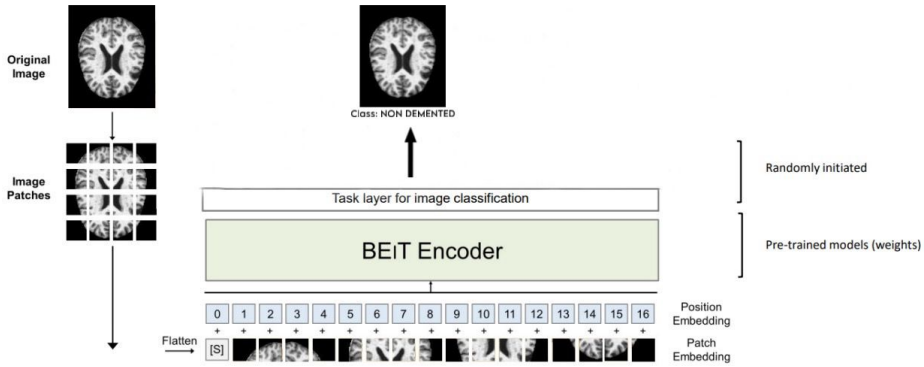


Figure 5.1: BeiT Architecture

5.4 TRAINING MODULE:

We introduce a task called masked image modeling (MIM), where we randomly obscure a portion of image patches and then predict the visual tokens corresponding to these masked patches.Given an input image, we divide it into several image patches and tokenize them into visual tokens. Around 40 percentage of these patches are randomly masked. These masked patches are then replaced with a learnable embedding. The resulting corrupted image patches are passed through a Transformer. The final hidden vectors serve as encoded representations of the input patches. For each masked position, we use a softmax classifier to predict the corresponding visual tokens. The objective of pre-training is to maximize the likelihood of predicting the correct visual tokens given the corrupted image.

This module's main emphasis is on training the deep learning model iteratively using the provided dataset. The training loop, which includes feeding the model data in batches for forward propagation, must be implemented at this stage. Throughout the forward propagation phase, the

model generates output probabilities for the several classes linked to the phases of Alzheimer's disease by computing predictions for each input sample. After forward propagation, the module uses the cross-entropy loss function to determine the loss that the model's predictions relative to the ground truth labels suffer. As a measure of the model's effectiveness during training, this loss metric measures the discrepancy between the anticipated and actual class distributions. The subsequent stage involves computing the gradients of the loss function concerning the model's parameters using backward propagation, commonly referred to as backpropagation. As a result of these gradients, the optimization process is guided by the direction and size of the modifications required to minimize the loss. Using the calculated gradients as input, the model parameters are updated using the Adam optimizer, a well-known variation of stochastic gradient descent (SGD). Adam optimizes and converges to an optimum solution efficiently by dynamically adjusting the learning rate for each parameter. The Training Module oversees the management of training hyperparameters such as learning rate, batch size, and the number of epochs. These parameters have a big impact on the training dynamics and the performance of the model. Finding the sweet spot between accurate models and efficient training requires fine-tuning these hyperparameters. To further evaluate the model's development and convergence, the Training Module allows for the tracking of critical performance indicators, such as accuracy and training loss. Researchers may improve model performance by regularly evaluating these indicators to detect any concerns like overfitting or underfitting and making the appropriate modifications. In order for the deep learning model to learn how to correctly categorize MRI pictures into various phases of Alzheimer's disease, the Training Module is crucial in directing the training

process. Scientists may improve the model's diagnostic accuracy and generalizability via repeated refinement by using the training loop, tweaking model parameters, and keeping an eye on performance indicators.

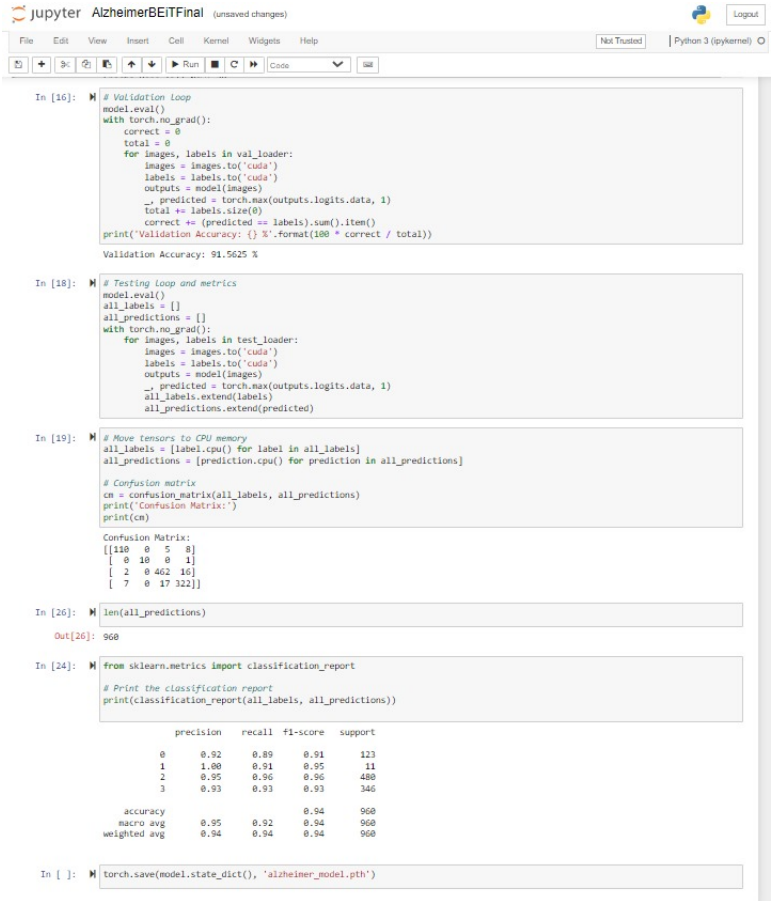
5.5 EVALUATION AND METRICS MODULE:

Assessing the trained model's performance on both the validation and test sets, as well as computing a variety of performance metrics necessary for assessing its classification skills, are the key objectives of the Evaluation and Metrics Module. This module's primary objective is to conduct a thorough quantitative analysis of the model's performance in order to draw conclusions about how well it performs when used in clinical settings to diagnose Alzheimer's disease. The lesson begins by calculating the most basic metrics, including recall, accuracy, precision, and F1-score. While accuracy evaluates how well the model performs in general, precision counts how many of its positive predictions actually came true. Recall, often called sensitivity, measures how well the model detected genuine positives relative to the total number of positive events. A fair evaluation of the model's efficacy in both categories may be found in the F1-score, which is a harmonic mean of recall and accuracy. Module output also includes a confusion matrix, a tabular summary of model predictions vs ground truth labels for various illness phases. By displaying regions of ambiguity or misclassification across categories, the confusion matrix permits an in-depth evaluation of the model's performance. Researchers may learn about the model's strengths and shortcomings by looking at the confusion matrix, which shows particular patterns of misclassification. If the trained model is to be used clinically for Alzheimer's disease diagnosis, the assessment and Metrics Module's assessment procedure is essential. The confusion matrix

and performance measures may be used to evaluate the model's generalizability, clinical value, and robustness. Clinical decision-making and the model's incorporation into diagnostic procedures are both greatly aided by this data. The Evaluation and Metrics Module is crucial since it gives a thorough evaluation of the trained model's performance, which sheds light on how well it worked and whether or not it would be a good fit for helping with clinical decision-making when diagnosing Alzheimer's disease. Researchers may ensure the model's diagnostic accuracy and make strides in medical imaging and deep learning-based diagnoses by conducting thorough evaluations and analyses.

Chapter 6

SOFTWARE TOOLS USED



jupyter AlzheimerBEiTFinal (unsaved changes)

Logout

File Edit View Insert Cell Kernel Widgets Help

Not Trusted | Python 3 (ipykernel)

Run Code

In [16]:
```
# Validation Loop
model.eval()
with torch.no_grad():
    correct = 0
    total = 0
    for images, labels in val_loader:
        images = images.to('cuda')
        labels = labels.to('cuda')
        outputs = model(images)
        _, predicted = torch.max(outputs.logits.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()
print('Validation Accuracy: {} %'.format(100 * correct / total))
```

```
Validation Accuracy: 91.5625 %
```

In [18]:
```
# Testing Loop and metrics
model.eval()
all_labels = []
all_predictions = []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to('cuda')
        labels = labels.to('cuda')
        outputs = model(images)
        _, predicted = torch.max(outputs.logits.data, 1)
        all_labels.extend(labels)
        all_predictions.extend(predicted)
```

In [19]:
```
# Move tensors to CPU memory
all_labels = [label.cpu() for label in all_labels]
all_predictions = [prediction.cpu() for prediction in all_predictions]

# Confusion matrix
cm = confusion_matrix(all_labels, all_predictions)
print('Confusion Matrix:')
print(cm)
```

```
Confusion Matrix:
[[110   0   5   8]
 [  0  10   0   1]
 [  2   0 462  16]
 [  7   0  17 322]]
```

In [26]:
```
len(all_predictions)
```

Out[26]:
```
960
```

In [24]:
```
from sklearn.metrics import classification_report

# Print the classification report
print(classification_report(all_labels, all_predictions))
```

```
              precision    recall  f1-score   support

           0       0.92      0.89      0.91       123
           1       1.00      0.91      0.95        11
           2       0.95      0.96      0.96       480
           3       0.93      0.93      0.93       346

    accuracy                           0.94       960
   macro avg       0.95      0.92      0.94       960
weighted avg       0.94      0.94      0.94       960
```

In [ ]:
```
torch.save(model.state_dict(), 'alzheimer_model.pth')
```

Figure 6.1: Python IDE

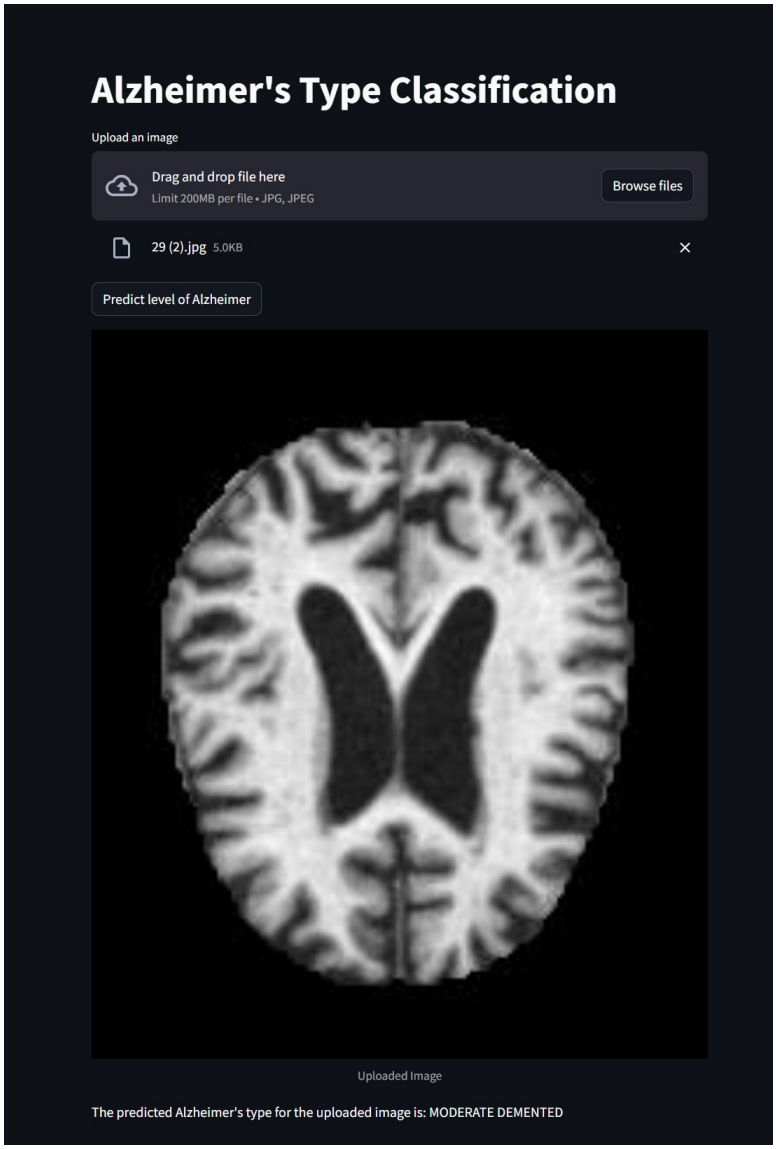


Figure 6.2: Prediction

Chapter 7

RESULTS & DISCUSSION

Model	Precision	Recall	F1-Score	Accuracy
BEiT (Proposed)	0.95	0.92	0.93	0.94
KNN	0.30	0.29	0.28	0.42
SVM	0.83	0.47	0.50	0.68
LDA	0.91	0.91	0.91	0.91
CNN	0.91	0.90	0.89	0.90
Inception Restnet	0.65	0.31	0.42	0.78
VGG	0.41	0.29	1.01	0.54
Inception V3	0.89	0.89	0.89	0.89
DenseNet	0.72	0.26	0.37	0.79

Table 7.1: Results

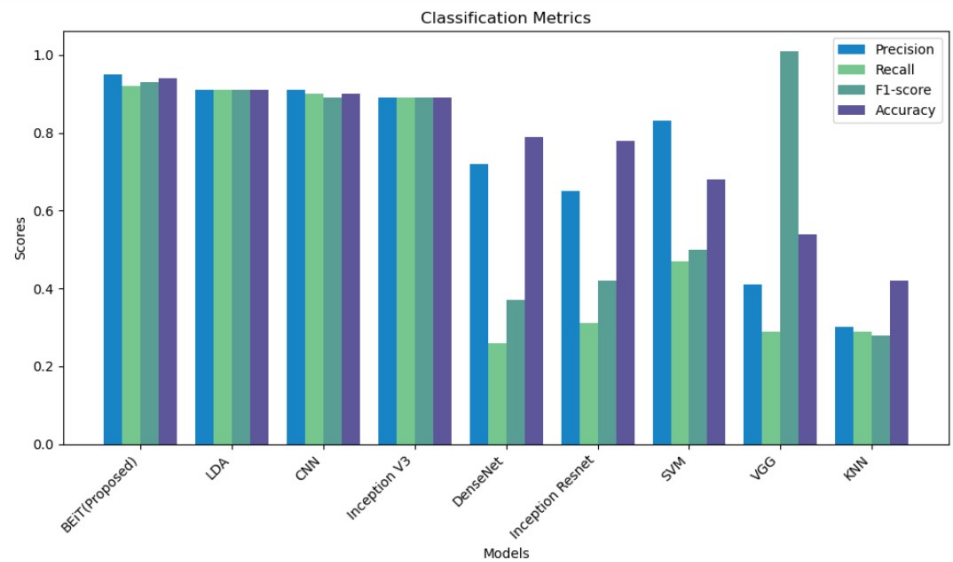


Figure 7.1: Performance Metrics Graph

In comparison to the other models tested for the project, the suggested BEiT model performs very well. Classifying MRI images for Alzheimer's

disease diagnosis, the BEiT model obtains an accuracy of 0.94 with a recall of 0.92, precision of 0.95, and F1-score of 0.93. In comparison to more conventional ML models like KNN, SVM, LDA, and CNN, as well as well-known DL architectures like Inception, ResNet, VGG, Inception V3, DenseNet, and EfficientNetB0, this demonstrates superior performance. Note that many models sacrifice performance in other measures in exchange for high recall or accuracy. For example, support vector machines (SVMs) have a respectable F1-score of 0.50 and an accuracy of 0.68 while having a poor recall of 0.47. Contrarily, the BEiT model exhibits resilience in correctly identifying MRI images throughout various stages of Alzheimer's disease, as it achieves a balanced performance across accuracy, recall, and F1-score. These findings demonstrate the promise of transformer-based designs such as BEiT for improving the accuracy of neurodegenerative disease diagnostics and their usefulness in medical picture processing.

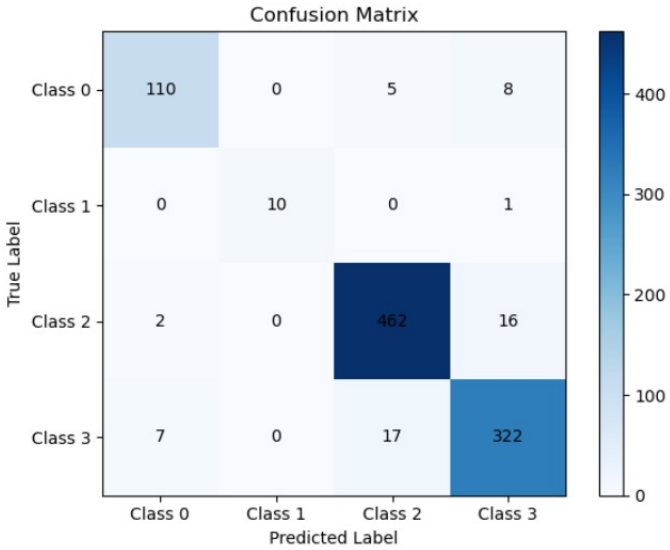


Figure 7.2: Confusion Matrix of BEiT

Chapter 8

CONCLUSION

In conclusion, this project demonstrates the effectiveness of utilizing transformer-based architectures, particularly the BEiT model, for the classification of MRI images in Alzheimer's disease diagnosis. The BEiT model showcases superior performance compared to traditional machine learning models and popular deep learning architectures, achieving a precision of 0.95, recall of 0.92, and F1-score of 0.93, resulting in an accuracy of 0.94. This indicates the robustness and accuracy of the BEiT model in accurately classifying MRI images across different classes of Alzheimer's disease. The project highlights the importance of utilizing advanced deep learning techniques in medical image analysis, offering potential avenues for enhancing diagnostic accuracy and facilitating early intervention in neurodegenerative disorders. By automating the process of image classification, the suggested model can aid healthcare professionals in making timely and informed decisions, ultimately improving patient care and outcomes. Overall, this project contributes to the advancement of AI-driven approaches in healthcare, with promising implications for neurodegenerative disease diagnosis and management. Furthermore, the successful deployment of transformers underscores their potential to address complex healthcare challenges, paving the way for innovative applications across various domains and underscoring their indispensable role in shaping the future of AI-driven healthcare.

8.1 SCOPE OF FUTURE WORK

In the scope of further exploration and improvement for Alzheimer's disease classification using the BEiT model, several key areas are identified. Firstly, expanding and augmenting the dataset with diverse samples and employing data augmentation techniques could bolster the model's generalization capabilities. Fine-tuning the pre-trained BEiT model on Alzheimer's disease-specific data sets, as well as exploring transfer learning from related tasks, offers avenues for improving classification performance. Moreover, developing interpretable models, such as attention mechanisms and saliency maps, could provide valuable insights into the underlying biological processes and aid clinicians in decision-making. Additionally, integrating information from multiple modalities, including MRI, PET scans, and clinical data, through fusion techniques may lead to more comprehensive diagnostic models. Conducting longitudinal studies to track disease progression over time and collaborating with healthcare professionals for clinical validation are essential steps towards real-world deployment. Ethical considerations, such as ensuring data privacy and addressing biases in the model, are paramount, and fostering open science practices can promote transparency and reproducibility in the field. Overall, a holistic approach that combines technical innovation with ethical and clinical considerations is crucial for advancing Alzheimer's disease diagnosis and treatment.

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